

DOC No. : S-1, GNEPIP Surajpur Industrial Area, Kasna Phase V,
Telephone : +91 9873571512 Greater Noida, Gautam Buddha Nagar, Gautam
FAX : - Buddha Nagar, Uttar Pradesh, India - 201308
E-Mail : qa@spectrogroup.com
BO Code : NA

Test REPORT AS PER : IS 16046 : Part 2 (2018)**QR Code/Barcode : 152922CRS****REPORT NO : SC23EPF10473_1**

DATE : 12 Jul, 2023

PART A. PARTICULARS OF SAMPLE SUBMITTED

a) Customer Name & Address : Xinxiang Hongli Supply Source Technology Co.,
Ltd
ZHONGDAYANG VILLAGE,ZHAIPO
TOWN,XINXIANG COUNTY, XINXIANG
CITY,HENAN PROVINCE, CHINA, NA, HENAN,
China - 453000

b) Nature of sample : -

c) Grade/Variety/Type/Class Size etc : NA

d) Declare values, if any : -

e) Batch No. & Date of Manufacture : /

f) Quantity : 45

g) Date of Receipt : 30 Jun, 2023

h) BIS Seal : Verified by Sample Cell

i) IO's Signature : Verified by Sample Cell

j) Any other Information / Expiry Date, If any : /

k) Date of Commencement of Testing : 30 Jun, 2023

l) Date of Completion of Testing : 12 Jul, 2023

m) Section Code : 23E833CN

n) Section Report No. : 23E833CN_1

o) Report Type : New

p) Reference Report No. :

q) Remarks :

Murad Alam
OIC SAMPLE CELL
(Authorized Signatory)
Authorized on: 12 Jul, 2023 14:48 PM

1.

This is a Computer Generated Report.

PART B. SUPPLEMENTARY INFORMATION

- | | |
|--|----------------|
| 1. Reference to sampling procedure, wherever applicable. | Not Applicable |
| 2. Supporting documents for the measurements taken and results derived like graphs, table sketches and or photographs as appropriate to test report, if any. | Yes |
| 3. Deviation from the test methods as prescribed in relevant ISS/Work instruction, if any. | Not Applicable |
| 3. NABL Report required ? | Yes |

SHIVENDRA KUMAR
OIC Electrical
(Authorized Signatory)
Authorized on: 12 Jul, 2023 14:46 PM

This is a Computer Generated Report.

PART C. TEST RESULT

S.No.	Clause No Table No. Sl. No	Parameter - Method of test	Test Description	Min Limit	Max Limit	Unit	Result/ Observation
1	10	Packaging	PACKAGING AND TRANSPORT	-	-	-	Attached in Pdf File
2	9.4	Other information	Storage and disposal instructions Recommended charging instructions	-	-	-	Test Not Applicable
3	9.3	Caution for ingestion of small cells and batteries	Caution for ingestion of small cells and batteries	-	-	-	Test Not Applicable
4	9.2	Battery Marking	Batteries marked as specified in IEC 61960, except for coin batteries	-	-	-	Test Not Applicable
5	9.1	Cell Marking	cell marking	-	-	-	Attached in Pdf File
6	9	Marking	marking	-	-	-	Attached in Pdf File
7	8.2	Small Cell and battery safety information	The Product is Small battery which was tested by injection gauge and observe it can fit within the limits of the ingestion gauge. Therefore, The following warning language is provided on packaging of Box. • Keep small cells and batteries which are considered swallowable out of the reach of children. • Swallowing may lead to burns, perforation of soft tissue, and death. Severe burns can occur within 2 h of ingestion. • In case of ingestion of a cell or battery, seek medical assistance promptly.	-	-	-	Test Not Applicable
8	8.1	General	N/A	-	-	-	Attached in Pdf File
9	8	Information for safety	N/A	-	-	-	Attached in Pdf File
10	7.3	Reasonably foreseeable misuse	Cl. 7.3.2, Cl. 7.3.3, Cl. 7.3.6, ,Cl.7.3.8.1,Cl. 7.3.8.2	-	-	-	Attached in Pdf File
11	7.2	Intended Use	7.2 Intended use 7.2.1 Continuous charging at constant voltage (cells) 7.2.2 Case stress at high ambient temperature (battery)	-	-	-	Attached in Pdf File
12	7.1	Charging procedure for test purposes	Charging procedure for test purposes	-	-	-	Attached in Pdf File
13	7	Specific requirements and tests	SPECIFIC REQUIREMENTS AND TESTS	-	-	-	Attached in Pdf File
14	6	Tyoe Test amd Sample Size	N/A	-	-	-	Attached in Pdf File

15	5.8	Battery Safety Components	N/A	-	-	-	Test Not Applicable
16	5.7	Quality Plan	Quality plan*	-	-	-	Attached in Pdf File
17	5.6	Assembly of Cells into batteries	5.6.1 General 5.6.2 Design recommendation 5.6.3 Mechanical protection for cells and components of batteries	-	-	-	Test Not Applicable
18	5.5	Terminal Contacts	Terminal contacts*	-	-	-	Attached in Pdf File
19	5.4	Temperature, Voltage and Current Management	Temperature, voltage and current management	-	-	-	Test Not Applicable
20	5.3	Venting	Venting*-Battery cases and cells incorporate a pressure relief mechanism or are constructed so that they relieve excessive internal pressure at a value and rate that will preclude rupture, explosion and self-ignition	-	-	-	Attached in Pdf File
21	5.2	Insulation and Wiring	Insulation and wiring	-	-	-	Test Not Applicable
22	5.1	General	Cells and batteries so designed and constructed that they are safe under conditions of both intended use and reasonably foreseeable misuse	-	-	-	Attached in Pdf File
23	5	General Safety Considerations	General	-	-	-	Attached In Pdf file
24	4	Parameter Measurement Tolerances	Parameter measurement tolerances	-	-	-	Attached in pdf file

SHIVENDRA KUMAR
OIC Electrical
 (Authorized Signatory)
 Authorized on: 12 Jul, 2023 14:46 PM

This is a Computer Generated Report.

.....

PART D. REMARKS

SHIVENDRA KUMAR
OIC Electrical
(Authorized Signatory)
Authorized on: 12 Jul, 2023 14:46 PM

This is a Computer Generated Report.



Discipline: Electrical
Group: Cells & batteries

ULR No. TC631423000007010F

SUMMARY OF TEST REPORT
TEST REPORT NO... SC23EPF10473_1...DATED 12/07/2023
(Number of pages in test report: page no. 1 to 40)

TEST FORMAT AS PER IS 16046 (Part 2):2018 / IEC 62133-2:2017

- Name of Manufacturer: **Xinxiang Hongli Supply Source Technology Co., Ltd.**
ZHONGDAYANG VILLAGE, ZHAIPO TOWN, XINXIANG COUNTY, XINXIANG CITY, HENAN PROVINCE, CHINA, 453000
- Item: **Rechargeable Li-ion Cell**
- Model: **Lead Model: IMR18650-2000mAh**
Series Models: IMR18650-800mAh, IMR18650-1200mAh, IMR18650-1300mAh, IMR18650-1500mAh
IMR18650-1800mAh, IMR14500-400mAh, IMR14500-500mAh, IMR14500-600mAh
- Model differences provided (if applicable): Yes
- Model differences verified as per MEITY Guidelines for series formulation: Yes
- Test Results:

SL. NO.	TEST REQUIREMENT	CLAUSE	VERDICT
1.	Parameters measurement tolerances	4.0	P
2.	Insulation and wiring	5.2	N/A
3.	Venting	5.3	P
4.	Temperature/Current management	5.4	N/A
5.	Terminal contacts	5.5	P
6.	Assembly of cells into Batteries	5.6	N/A
7.	Quality Plan	5.7	P
8.	Battery safety components	5.8	N/A
9.	Type test conditions	6.0	P
10.	Charging procedure for test purpose	7.1	P
11.	Intended use	7.2	P
12.	Reasonably foreseeable misuse	7.3	P
13.	Information of safety	8.0	P
14.	Marking	9.1	P
15.	Packaging	10.0	P
16.	Charging range of secondary lithium ion cells for safe use	Annex A	P
17.	Measurement of the internal AC resistance for coin cells	Annex D	N/A

General Information:

1.The conformity certificates of critical components are verified to ensure complete testing of apparatus under test and details regarding harmonized IEC standards(where IEC standard are not available) are also provided in the list of critical component.

CONCLUSION :

- Sample meets all relevant requirements of **IS 16046 (Part 2):2018 / IEC 62133-2:2017.** ✓

Remarks:

- P:Pass; F: Fail; N/A: Not applicable**

I, hereby, undertake that the verdict stated in the test reports for all the tests matches with the test results. The sample meets all relevant requirements of IS 16046 (Part 2):2018 / IEC 62133-2:2017.If any deviation is found, suitable punitive action may be taken by BIS.


Tested By


Authorized Signatory
Sudhakar Kumar
Senior Analyst
Spectro Analytical Labs Private Limited
Greater Noida



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 1 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Manufacturer:	Xinxiang Hongli Supply Source Technology Co., Ltd. ZHONGDAYANG VILLAGE, ZHAIPO TOWN, XINXIANG COUNTY, XINXIANG CITY, HENAN PROVINCE, CHINA, 453000		
Test item:	Rechargeable Li-ion Cell		
Identification:	Lead Model: IMR18650-2000mAh Series Model: IMR18650-800mAh, IMR18650-1200mAh IMR18650-1300mAh, IMR18650-1500mAh IMR18650-1800mAh, IMR14500-400mAh, IMR14500-500mAh, IMR14500-600mAh	Serial No.:	N/A
Lab Reference No.:	N230630001-1	Date of receipt:	30/06/2023
Testing laboratory and its address:	SPECTRO ANALYTICAL LABS PRIVATE LIMITED S-1, GNEPIP, SURAJPUR INDUSTRIAL AREA PHASE-5, KASNA, GREATER NOIDA-201308		
Test specification:	IS 16046 (Part 2):2018 / IEC 62133-2:2017		
Test Result:	The test item passed the test specification(s).		
Other Aspects:	1. P = Pass, F = Fail, N/A = Not Applicable 2. Model: IMR18650-800mAh, IMR18650-1200mAh, IMR18650-1300mAh, IMR18650-1500mAh, IMR18650-1800mAh, IMR14500-400mAh, IMR14500-500mAh, IMR14500-600mAh are in the same product families and can be undertaken in series with model: IMR18650-2000mAh. All the tests have been performed on model: IMR18650-2000mAh (Worst case).		
This test report relates to the test sample submitted and list of documents attached.			

Tested by:	Approved by / Authorized Signatory:	Issued by:
(Rahul Kumar/Engineer)	(Shivendra kumar/Sr. Analyst)	(Shivendra kumar/Sr. Analyst)
Dated: 12/07/2023	Dated: 12/07/2023	Dated: 12/07/2023



TC-6314

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 2 of 40

Dated:12/07/2023

IEC 62133-2:2017

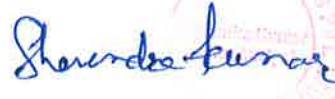
ULR No TC631423000007010F

TEST REPORT

IS 16046 (Part 2):2018 / IEC 62133-2:2017

Secondary cells and batteries containing alkaline or other non-acid electrolytes – Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications -
Part 2: Lithium systems

Total number of pages	40
Testing Laboratory	SPECTRO ANALYTICAL LABS PRIVATE LIMITED
Address	S-1, GNEPIP, SURAJPUR INDUSTRIAL AREA PHASE-5, KASNA, GREATER NOIDA-201308
Manufacturer's name	Xinxiang Hongli Supply Source Technology Co., Ltd.
Address	ZHONGDAYANG VILLAGE, ZHAIPO TOWN, XINXIANG COUNTY, XINXIANG CITY, HENAN PROVINCE, CHINA, 453000
Test specification:	
Standard	IS 16046 (Part 2):2018 / IEC 62133-2:2017
Test procedure	BIS Compliance Report
Non-standard test method.....:	N/A
Test Report Form No.:	BIS_BAT/SCAB_IS16046(PART2)_V1.0
Test Report Form(s) Originator	Bureau of Indian Standards
Master TRF	10.01.2019
Test item description	Rechargeable Li-ion Cell
Brand Name	
Model/Type reference	Lead Model: IMR18650-2000mAh Series Model: IMR18650-800mAh, IMR18650-1200mAh IMR18650-1300mAh, IMR18650-1500mAh IMR18650-1800mAh, IMR14500-400mAh, IMR14500-500mAh, IMR14500-600mAh
Ratings.....:	3.7V, 2000mAh
Other Documents submitted	Please refer to Table – List of Attachments at Page No. 35

Tested by:	Approved by / Authorized Signatory:	Issued by:
		
(Rahul Kumar/Engineer)	(Shivendra kumar/Sr. Analyst)	(Shivendra kumar/Sr. Analyst)
Dated: 12/07/2023	Dated: 12/07/2023	Dated: 12/07/2023

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 3 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Description	Measurement/ testing	Total No. of tests	Total no. of applicable tests/ Req.	No. of tests/ Req. passed	Page No.
General Requirements	Parameter measurement tolerances (Cl. 4)	01	01	01	10
General safety considerations	Insulation and wiring (Cl. 5.2)	09	00	N/A	11
General safety considerations	Venting (Cl. 5.3)	03	02	02	12
General safety considerations	Temperature/voltage/Current management (Cl. 5.4)	04	00	N/A	13
General safety considerations	Terminal contacts (Cl. 5.5)	04	04	04	14
General safety considerations	Assembly of cells into batteries (Cl.5.6)	23	00	N/A	15
General safety considerations	Quality plan (Cl. 5.7)	02	02	02	17
General safety considerations	Battery safety components	02	00	N/A	18
Type test and sample size	Type test conditions (Cl. 6.0)	05	02	02	19
Specific requirements and tests	Charging procedure for test purposes (Cl. 7.1)	08	07	07	20
Specific requirements and tests	Intended use (Cl. 7.2)	07	04	04	21
Specific requirements and tests	Reasonably foreseeable misuse (Cl. 7.3)	47	20	17	22
General Requirements	Information for Safety (Cl. 8)	11	02	02	25
Marking Requirements	Marking (Cl. 9.1)	15	02	02	26
Mechanical properties	Packaging (Cl. 10)	02	01	01	28
Charging and discharging range of secondary lithium ion cells for safe use	Charging range of secondary lithium ion cells for safe use(Annex A)	50	29	29	29
Measurement of the internal AC resistance for coin cells	Measurement of the internal AC resistance for coin cells(Annex D)	05	00	N/A	33

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested..

[Signature]
(Approving Authority)

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 4 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Table-List of Attachments

Attachment No.	Attachment Description	No. of pages in Attachment
Attachment – 1	Photo documents	01

Shivendra Kumar
Shivendra Kumar
Senior Analyst



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 5 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Copy of Marking label:

Lead Model: IMR18650-2000mAh

Rechargeable Li-ion Cell IMR18650-2000mAh

+ 3.7V 2000mAh 7.4Wh IMR19/66 -

2023.6.15 Made In China



XinXiang Hongli Supply Source Technology Co.,Ltd

Series Model: IMR18650-800mAh

Rechargeable Li-ion Cell IMR18650-800mAh

+ 3.7V 800mAh 2.96Wh IMR19/66 -

2023.6.15 Made In China



XinXiang Hongli Supply Source Technology Co.,Ltd

Series Model: IMR18650-1200mAh

Rechargeable Li-ion Cell IMR18650-1200mAh

+ 3.7V 1200mAh 4.44Wh IMR19/66 -

2023.6.15 Made In China



XinXiang Hongli Supply Source Technology Co.,Ltd



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 6 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Series Model: IMR18650-1300mAh

Rechargeable Li-ion Cell IMR18650-1300mAh

+ 3.7V 1300mAh 4.81Wh IMR19/66 -

2023.6.15 Made In China

XinXiang Hongli Supply Source Technology Co.,Ltd



Series Model: IMR18650-1500mAh

Rechargeable Li-ion Cell IMR18650-1500mAh

+ 3.7V 1500mAh 5.55Wh IMR19/66 -

2023.6.15 Made In China

XinXiang Hongli Supply Source Technology Co.,Ltd



Series Model: IMR18650-1800mAh

Rechargeable Li-ion Cell IMR18650-1800mAh

+ 3.7V 1800mAh 6.66Wh IMR19/66 -

2023.6.15 Made In China

XinXiang Hongli Supply Source Technology Co.,Ltd





TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 7 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Series Model: IMR14500-400mAh

Rechargeable Li-ion Cell IMR14500-400mAh

+ 3.7V 400mAh 1.48Wh IMR15/51 -



2023.6.15 Made In China

XinXiang Hongli Supply Source Technology Co.,Ltd

Series Model: IMR14500-500mAh

Rechargeable Li-ion Cell IMR14500-500mAh

+ 3.7V 500mAh 1.85Wh IMR15/51 -



2023.6.15 Made In China

XinXiang Hongli Supply Source Technology Co.,Ltd

Series Model: IMR14500-600mAh

Rechargeable Li-ion Cell IMR14500-600mAh

+ 3.7V 600mAh 2.22Wh IMR15/51 -



2023.6.15 Made In China

XinXiang Hongli Supply Source Technology Co.,Ltd

Shivendra Kumar
Spectro Analytical Labs Pvt. Ltd.
Sr. Analyst



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 8 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Test Items particulars..... :	Rechargeable Li-ion Cell
Classification of installation and use..... :	Class III & used in the portable application
Supply connection..... :	Not directly connected to mains
Recommended charging method declared by the manufacturer..... :	CC/CV
Discharge current(0.2ItmA)..... :	400mA
Specified final charge voltage..... :	4.2V
Specified charging rate..... :	400mA
Specified final discharge voltage..... :	3.0V
Upper limit charging voltage per cell..... :	4.2V
Maximum charging current..... :	1000mA
Charging temperature upper limit..... :	45°C
Charging temperature lower limit	0°C
Polymer cell electrolyte type..... :	☐ gel polymer..... ☐ solid polymer..... ☒ N/A.....
Possible test case verdicts:	
test case does not apply to the test object..... :	N/A
- test object does meet the requirement..... :	P (Pass)
- test object does not meet the requirement..... :	F (Fail)
Testing :	
Date of receipt of test item..... :	30/06/2023
Date (s) of performance of tests..... :	30/06/2023 to 12/07/2023
General remarks:	
The test results presented in this report relate only to the object tested. This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory. Throughout this report a “,” comma / “.” point is used as the decimal separator.	
Laboratory Conditions	
Ambient Temperature	(20 ± 5)°C
Ambient Humidity	(60 ± 15)% RH

Shraddha Kumar
SPECTRO ANALYTICAL LABS PRIVATE LIMITED

BIS_BAT/SCAB_IS16046(PART2)_V1.0

S-1, GNEPIP, Surajpur Industrial Area, Kaska, Greater Noida, Phase V, Gautam Budha Nagar (U.P) 201308



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 9 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

General product information:

1) Application details / Description of the product:

Product Under Test is "Rechargeable Li-ion Cell "

Brand Name



Model

Lead Model: IMR18650-2000mAh
Series Model: IMR18650-800mAh, IMR18650-1200mAh
IMR18650-1300mAh, IMR18650-1500mAh
IMR18650-1800mAh, IMR14500-400mAh,
IMR14500-500mAh, IMR14500-600mAh

Rating

: 3.7V,2000mAh

Charging specification:

Model	Upper Limited Charging Voltage (Vdc)	Standard Charging Current (mA)	Maximum Charging Current (mA)	Standard Discharging Current (0.2It mA)	End Discharge Voltage (Vdc)
IMR18650-2000mAh	4.2	400	1000	400	3.0

Max. specified ambient temperature (°C)..... :

Charging Temperature(0°C~45°C), Discharging Temperature(-10°C~60°C)

2) Differences between the models:

Model:	Capacity(mAh)	Height(mm)	Diameter(mm)	Weight (g)
IMR18650-2000mAh	2000	66	18.8	43
IMR14500-400mAh	400	51	14.2	14.5
IMR14500-500mAh	500	51	14.2	16.5
IMR14500-600mAh	600	51	14.2	18
IMR18650-800mAh	800	66	19	27
IMR18650-1200mAh	1200	66	18.5	36
IMR18650-1300mAh	1300	66	18.5	37
IMR18650-1500mAh	1500	66	19	40
IMR18650-1800mAh	1800	66	19	43

3) Similarities: Yes

Same nominal voltage.
Cells of same construction design.
Same Electrodes /Electrolytes.

4) Model No. tested with-in the family series....

IMR18650-2000mAh (Worst Case)

5) Options: N/A

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1	IS 16046 (Part 2):2018	Page 10 of 40
Dated:12/07/2023	IEC 62133-2:2017	ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
4	PARAMETER MEASUREMENT TOLERANCES		P
	Parameter measurement tolerances	All controlled and measured values were within the tolerances.	P

Total number of Requirements to be observed / inspected = 01

Total No of applicable Requirement = 01

No of Requirements for which the sample passed = 01

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.


(Approving Authority)



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 11 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

5	GENERAL SAFETY CONSIDERATIONS		P
5.1	General	The Cell is safe and Continue to function in all respect of its intended use, the Cell is safe and does not present significant Hazards under the condition of reasonably foreseeable misuse	P
	Cells and batteries so designed and constructed that they are safe under conditions of both intended use and reasonably foreseeable misuse	Complied	P
5.2	Insulation and Wiring	See below	N/A
	The insulation resistance between the positive terminal and externally exposed metal surfaces of the battery (excluding electrical contact surfaces) is not less than 5 MΩ.	Considered for battery Pack only	N/A
	Insulation resistance (MΩ):		N/A
	Internal wiring and its insulation are sufficient to withstand maximum anticipated current, voltage and temperature requirements		N/A
	Orientation of wiring maintains adequate creepage and clearance distances between connectors		N/A
	Mechanical integrity of internal connections accommodates conditions of reasonably foreseeable misuse		N/A

Total number of Requirements to be observed / inspected = 02

Total No of applicable Requirement = 02

No of Requirements for which the sample passed = 02

Total number of tests to be conducted = 07

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.

(Approving Authority)



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1	IS 16046 (Part 2):2018	Page 12 of 40
Dated:12/07/2023	IEC 62133-2:2017	ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

5.3	Venting	See below	P
	Battery cases and cells incorporate a pressure relief mechanism or are constructed so that they relieve excessive internal pressure at a value and rate that will preclude rupture, explosion and self-ignition	The edge of packing which next to terminal was considered as the pressure relief mechanism, which can release pressure during the abnormal condition	P
	Encapsulation used to support cells within an outer casing neither cause the battery to overheat during normal operation nor inhibit pressure relief	The cell is a built in product, which shall be enclosed in a rigid case, and will be evaluated in a final system	N/A

Total number of Requirements to be observed / inspected = 01
Total No of applicable Requirement = 01
No of Requirements for which the sample passed = 01

Total number of tests to be conducted = 02
Total No of applicable Tests = 01
No. of tests for which the sample passed = 01

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 13 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

5.4	Temperature, voltage and current management	Considered for battery pack only	N/A
	Batteries are designed such that abnormal temperature rise conditions are prevented	As above	N/A
	Batteries are designed to be within temperature, voltage and current limits specified by the cell manufacturer	As above	N/A
	Batteries are provided with specifications and charging instructions for equipment manufacturers so that associated chargers are designed to maintain charging within the temperature, voltage and current limits specified	As above	N/A

Total number of Requirements to be observed / inspected = 00

Total No of applicable Requirement = 00

No of Requirements for which the sample passed = N/A

Total number of tests to be conducted = 04

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)

BIS_BAT/SCAB_IS16046(PART2)_V1.0

S-1, GNEPIP, Surajpur Industrial Area, Kasna, Greater Noida, Phase V, Gautam Budha Nagar (U.P) 201308



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 14 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

5.5	Terminal Contacts	See below	P
	The size and shape of the terminal contacts ensure that they can carry the maximum anticipated current.	Complied	P
	External terminal contact surfaces are formed from conductive materials with good mechanical strength and corrosion resistance.	Complied	P
	Terminal contacts are arranged to minimize the risk of short circuits.	Satisfactory	P

Total number of Requirements to be observed / inspected = 04

Total No of applicable Requirement = 04

No of Requirements for which the sample passed = 04

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.

(Approving Authority)



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1	IS 16046 (Part 2):2018	Page 15 of 40
Dated:12/07/2023	IEC 62133-2:2017	ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

5.6	Assembly of cells into batteries	Considered for battery pack only	N/A
5.6.1	General	As above	N/A
	Each battery have an independent control and protection for current, voltage, temperature and any other parameter required for safety and to maintain the cells within their operating region	As above	N/A
	This protection may be provided external to the battery such as within the charger or the end devices	As above	N/A
	If protection is external to the battery, the manufacturer of the battery provide this safety relevant information to the external device manufacturer for implementation	As above	N/A
	If there is more than one battery housed in a single battery case, each battery have protective circuitry that can maintain the cells within their operating regions	As above	N/A
	Manufacturers of cells specify current, voltage and temperature limits so that the battery manufacturer/designer may ensure proper design and assembly	As above	N/A
	Batteries that are designed for the selective discharge of a portion of their series connected cells incorporate circuitry to prevent operation of cells outside the limits specified by the cell manufacturer	As above	N/A
	Protective circuit components added as appropriate and consideration given to the end-device application	As above	N/A
	The manufacturer of the battery provide a safety analysis of the battery safety circuitry with a test report including a fault analysis of the protection circuit under both charging and discharging conditions confirming the compliance	As above	N/A
5.6.2	Design recommendation	As above	N/A
	For the battery consisting of a single cell or a single cellblock, it is recommended that the charging voltage of the cell does not exceed the upper limit of the charging voltage specified in Table 2	As above	N/A
	For the battery consisting of series-connected plural single cells or series-connected plural cellblocks, it is recommended that the voltages of any one of the single cells or single cellblocks does not exceed the upper limit of the charging voltage, specified in Table 2, by monitoring the voltage of every single cell or the single cellblocks	As above	N/A
	For the battery consisting of series-connected plural single cells or series-connected plural cellblocks, it is recommended that charging is stopped when the upper limit of the charging voltage is exceeded for any one of the single cells or single cellblocks by measuring the voltage of every single cell or the single cellblocks	As above	N/A



Shivendra Kumar
Senior Analyst

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 16 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

	For batteries consisting of series-connected cells or cell blocks, nominal charge voltage not be counted as an overcharge protection	As above	N/A
	For batteries consisting of series-connected cells or cell blocks, cells have closely matched capacities, be of the same design, be of the same chemistry and be from the same manufacturer	As above	N/A
	It is recommended that the cells and cell blocks not discharged beyond the cell manufacturer's specified final voltage	As above	N/A
	For batteries consisting of series-connected cells or cell blocks, cell balancing circuitry incorporated into the battery management system	As above	N/A
5.6.3	Mechanical protection for cells and components of batteries	As above	N/A
	Mechanical protection for cells, cell connections and control circuits within the battery provided to prevent damage as a result of intended use and reasonably foreseeable misuse	As above	N/A
	The mechanical protection can be provided by the battery case or it can be provided by the end product enclosure for those batteries intended for building into an end product	As above	N/A
	The battery case and compartments housing cells designed to accommodate cell dimensional tolerances during charging and discharging as recommended by the cell manufacturer	As above	N/A
	For batteries intended for building into a portable end product, testing with the battery installed within the end product considered when conducting mechanical tests	As above	N/A

Total number of Requirements to be observed / inspected = 23

Total No of applicable Requirement = 00

No of Requirements for which the sample passed = N/A

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 17 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

5.7	Quality Plan	See below	P
	The manufacturer prepares and implements a quality plan that defines procedures for the inspection of materials, components, cells and batteries and which covers the whole process of producing each type of cell or battery.	ISO 9001:2015 provided	P

Total number of Requirements to be observed / inspected = 02

Total No of applicable Requirement = 02

No of Requirements for which the sample passed = 02

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1	IS 16046 (Part 2):2018	Page 18 of 40
Dated:12/07/2023	IEC 62133-2:2017	ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

5.8	Battery safety components	Considered for battery pack only	N/A
	According annex F	As above	N/A

Total number of Requirements to be observed / inspected = 02
Total No of applicable Requirement = 00
No of Requirements for which the sample passed = N/A

Total number of tests to be conducted = 00
Total No of applicable Tests = 00
No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.


(Approving Authority)





TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 19 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

6	TYPE TEST AND SAMPLE SIZE		P
	Tests are made with the number of cells or batteries specified in Table 1 using cells or batteries that are not more than six months old	Complied	P
	Coin cells with resistance $\leq 3 \Omega$ (measured according annex D) are tested according table 1	This is not a coin cell	N/A
	Unless otherwise specified, tests are carried out in an ambient temperature of $20^\circ\text{C} \pm 5^\circ\text{C}$	Considered and evaluated for the Cells	P
	The safety analysis of 5.6.1 identify those components of the protection circuit that are critical for short-circuit, overcharge and overdischarge protection	Considered for battery pack only	N/A
	When conducting the short-circuit test, consideration given to the simulation of any single fault condition that is likely to occur in the protecting circuit that would affect the short-circuit test	As above	N/A

Total number of Requirements to be observed / inspected = 05

Total No of applicable Requirement = 02

No of Requirements for which the sample passed = 02

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)

BIS_BAT/SCAB_IS16046(PART2)_V1.0

S-1, GNEPIP, Surajpur Industrial Area, Kasna, Greater Noida, Phase V, Gautam Budha Nagar (U.P) 201308

+91-11-4052-2000, +91-11-4161-1000 | www.spectro.in | care@spectro.in



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 20 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

7	SPECIFIC REQUIREMENTS AND TESTS		P
7.1	Charging procedure for test purposes	In compliance	P
7.1.1	First procedure		P
	This charging procedure applies to subclauses other than those specified in 7.1.2	Except Subclause specified in Cl.no. 7.1.2, First procedure used	P
	Unless otherwise stated in this document, the charging procedure for test purposes is carried out in an ambient temperature of 20 °C ± 5 °C, using the method declared by the manufacturer	Complied	P
	Prior to charging, the battery have been discharged at 20 °C ± 5 °C at a constant current of 0,2 It A down to a specified final voltage	Considered for battery pack	N/A
7.1.2	Second procedure	Second Procedure used for test of Clause No. 7.3.1, 7.3.4 and 7.3.5	P
	This charging procedure applies only to 7.3.1, 7.3.4, 7.3.5, and 7.3.9	Considered (See annex A)	P
	After stabilization for 1 h and 4 h, respectively, at ambient temperature of highest test temperature and lowest test temperature, as specified in Table 2, cells are charged by using the upper limit charging voltage and maximum charging current, until the charging current is reduced to 0,05 It A, using a constant voltage charging method	Complies	P

Total number of Requirements to be observed / inspected = 00

Total No of applicable Requirement = 00

No of Requirements for which the sample passed = N/A

Total number of tests to be conducted = 08

Total No of applicable Tests = 07

No. of tests for which the sample passed = 07

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.

(Approving Authority)

BIS_BAT/SCAB_IS16046(PART2)_V1.0

S-1, GNEPIP, Surajpur Industrial Area, Kasna, Greater Noida, Phase V, Gautam Budha Nagar (U.P) 201308



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 21 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
7.2	Intended Use	See below	P
7.2.1	Continuous charging at constant voltage (cells)	See below	P
	Fully charged cells are subjected for 7 days to a charge using the charging method for current and standard voltage specified by the cell manufacturer.	Five fully charged cells are subjected for 7days to a charge as specified by manufacturer	P
	Results: No fire. No explosion. No leakage..... :	No Fire, No Explosion, No Leakage during and after the test (See appended table 7.2.1)	P
7.2.2	Case stress at high ambient temperature (battery)	Considered for battery pack only	N/A
	Oven temperature (°C)	As above	N/A
	Results: No physical distortion of the battery case resulting in exposure of internal protective components and cells.	As above	N/A

Total number of Requirements to be observed / inspected = 05

Total No of applicable Requirement = 03

No of Requirements for which the sample passed = 03

Total number of tests to be conducted = 02

Total No of applicable Tests = 01

No. of tests for which the sample passed = 01

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.

Shruti Kumar
.....
(Approving Authority)

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 22 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
7.3	Reasonably foreseeable misuse	See below	P
7.3.1	External short circuit (cell)		P
	The cells were tested until one of the following occurred:		P
	- 24 hours elapsed; or	See below	N/A
	- The case temperature declined by 20 % of the maximum temperature rise	complied	P
	Results: No fire. No explosion	no fire, no explosion (See appended table 7.3.1)	P
7.3.2	External short circuit (battery)	Considered for battery only	N/A
	The batteries were tested until one of the following occurred:	As above	N/A
	- 24 hours elapsed; or	As above	N/A
	- The case temperature declined by 20 % of the maximum temperature rise	As above	N/A
	In case of rapid decline in short circuit current, the battery pack remained on test for an additional one hour after the current reached a low end steady state condition	As above	N/A
	A single fault in the discharge protection circuit conducted on one to four (depending upon the protection circuit) of the five samples before conducting the short-circuit test	As above	N/A
	A single fault applies to protective component parts such as MOSFET, fuse, thermostat or positive temperature coefficient (PTC) thermistor	As above	N/A
	Results: No fire. No explosion	As above	N/A
7.3.3	Free fall	See below	P
	Results: No fire. No explosion	No fire, No explosion occurred	P
7.3.4	Thermal abuse (cells)	See below	P
	Oven temperature (°C)	130°C±2°C	P
	Results: No fire. No explosion	No fire, no explosion after the test	P
7.3.5	Crush (cells)	See Below	P
	The crushing force was released upon:		P

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 23 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
	- The maximum force of 13 kN±0.78 kN has been applied; or	Maximum force of 13 kN±0.78 kN has been applied (See appended table 7.3.5)	P
	- An abrupt voltage drop of one-third of the original voltage has been obtained	No such condition observed	N/A
	Results: No fire. No explosion	No fire, no explosion occurred after the test (See appended table 7.3.5)	P
7.3.6	Over-charging of battery	Considered for battery pack only	N/A
	The supply voltage which is:	As above	N/A
	- 1.4 times the upper limit charging voltage presented in Table A.1 (but not to exceed 6.0 V) for single cell/cell block batteries or	As above	N/A
	- 1.2 times the upper limit charging voltage resented in Table A.1 per cell for series connected multi-cell batteries, and	As above	N/A
	- Sufficient to maintain a current of 2.0 It A throughout the duration of the test or until the supply voltage is reached.	As above	N/A
	Test was continued until the temperature of the outer casing:	As above	N/A
	- Reached steady state conditions (less than 10 °C change in 30-minute period); or	As above	N/A
	- Returned to ambient	As above	N/A
	Results: No fire. No explosion	As above	N/A
7.3.7	Forced discharge (cells)	See below	P
	If the discharge voltage reaches the negative value of upper limit charging voltage within the testing duration, the voltage is maintained at the negative value of the upper limit charging voltage by reducing the current for the remainder of the testing duration	Complied	P
	If the discharge voltage does not reach the negative value of upper limit charging voltage within the testing duration, the test is terminated at the end of the testing duration	No such condition observed	N/A
	Results: No fire. No explosion	No fire , no explosion caused after the test (See appended table 7.3.7)	P
7.3.8	Mechanical tests (batteries)	Considered for battery pack only	N/A
7.3.8.1	Vibration	As above	N/A
	Results: No fire, no explosion, no rupture, no leakage or venting.	As above	N/A

BIS_BAT/SCAB_IS16046(PART2)_V1.0

Shwendra Kumar
Senior Analyst



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 24 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
7.3.8.2	Mechanical shock	As above	N/A
	Results: No leakage, no venting, no rupture, no explosion and no fire	As above	N/A
7.3.9	Design evaluation – Forced internal short circuit (cells)		N/A
	The cells complied with national requirement for	This Country specific test is not applicable in India	N/A
	The pressing was stopped upon:		N/A
	- A voltage drop of 50 mV has been detected; or		N/A
	- The pressing force of 800 N (cylindrical cells) or 400 N (prismatic cells) has been reached		N/A
	Results: No fire.....		N/A

Total number of Requirements to be observed / inspected = 12

Total No of applicable Requirement = 07

No of Requirements for which the sample passed = 07

Total number of tests to be conducted = 35

Total No of applicable Tests = 10

No. of tests for which the sample passed = 10

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 25 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
8	INFORMATION FOR SAFETY		P
8.1	General	See below	P
	Manufacturers of secondary cells ensure that information is provided about current, voltage and temperature limits of their products	Current, voltage and temperature limits are provided In manufacturer specification	p
	Manufacturers of batteries ensure that equipment manufacturers and, in the case of direct sales, end-users are provided with information to minimize and mitigate hazards	Considered in battery pack only	N/A
	Systems analyses performed by device manufacturers to ensure that a particular battery design prevents hazards from occurring during use of a product	As above	N/A
	As appropriate, any information relating to hazard avoidance resulting from a system analysis provided to the end user	As above	N/A
	Do not allow children to replace batteries without adult supervision	As above	N/A
8.2	Small cell and battery safety information	See below	N/A
	The following warning language is to be provided with the information packaged with the small cells and batteries or equipment using them:	This is not small cell	N/A
	-Keep small cells and batteries which are considered swallowable out of the reach of children.	As above	N/A
	-Swallowing may lead to burns, perforation of soft tissue, and death. Severe burns can occur within 2h of ingestion.	As above	N/A
	-In case of ingestion of a cell or battery, seek medical assistance promptly.	As above	N/A

Total number of Requirements to be observed / inspected = 11

Total No of applicable Requirement = 02

No of Requirements for which the sample passed = 02

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.

(Approving Authority)

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 26 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
9	Markings		P
9.1	Cell marking	See below	P
	Cells marked as specified in IEC 61960, except coin cells	Cell used for manufacturing of battery	N/A
	Coin cells whose external surface area is too small to accommodate the markings on the cells show the designation and polarity	This is not a coin cell	N/A
	By agreement between the cell manufacturer and the battery and/or end product manufacturer, component cells used in the manufacture of a battery need not be marked		N/A
9.2	Battery marking	See below	N/A
	Batteries marked as specified in IEC 61960, except for coin batteries	Product is Rechargeable Li-ion Cell	N/A
	Coin batteries whose external surface area is too small to accommodate the markings on the batteries show the designation and polarity. Batteries also marked with an appropriate caution statement	As above	N/A
	Terminals have clear polarity marking on the external surface of the battery	As above	N/A
	Batteries with keyed external connectors designed for connection to specific end products need not be marked with polarity markings if the design of the external connector prevents reverse polarity connections	As above	N/A
9.3	Caution for ingestion of small cells and batteries	See below	N/A
	Coin cells and batteries identified as small batteries according to 8.2 include a caution statement regarding the hazards of ingestion in accordance with 8.2.	This is not a coin cell	N/A
	When small cells and batteries are intended for direct sale in consumer-replaceable applications, caution for ingestion given on the immediate package.	This is not a small cell	N/A
9.4	Other Information	See below	N/A
	Storage and disposal instructions	This is not a battery	N/A
	Recommended charging instructions	As above	N/A





TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 27 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Total number of Requirements to be observed / inspected = 15

Total No of applicable Requirement = 02

No of Requirements for which the sample passed = 02

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.


Shwendra Kumar
Senior Analyst

(Approving Authority)



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1	IS 16046 (Part 2):2018	Page 28 of 40
Dated:12/07/2023	IEC 62133-2:2017	ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
10	PACKAGING AND TRANSPORT		P
	Packaging for coin cells not small enough to fit within the limits of the ingestion gauge of Figure 3	This is not a coin cell	N/A
	The materials and packaging design are chosen so as to prevent the development of unintentional electrical conduction, corrosion of the terminals and ingress of environmental contaminants	Satisfactory	P

Total number of Requirements to be observed / inspected = 02

Total No of applicable Requirement = 01

No of Requirements for which the sample passed = 01

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 29 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
ANNEX A	CHARGING AND DISCHARGING RANGE OF SECONDARY LITHIUM ION CELLS FOR SAFE USE		P
A.1	General	See below	P
A.2	Safety of lithium-ion secondary battery		P
A.3	Consideration on charging voltage	See below	P
A.3.1	General	Charging voltage is not exceeded the value which is specified by the manufacturer	P
A.3.2	Upper limit charging voltage	See below	P
A.3.2.1	General	Upper limit charging voltage is 4.2V	P
A.3.2.2	Explanation of safety viewpoint	Charging voltage is 4.2V	P
A.3.2.3	Safety requirements, when different upper limit charging voltage is applied	Considered	P
A.4	Consideration of temperature and charging current	In compliance	P
A.4.1	General	See below	P
A.4.2	Recommended temperature range	Charge temperature declared by the manufacture 0~45°C	P
A.4.2.1	General	See below	P
A.4.2.2	Safety consideration when a different recommended temperature range is applied	Test carried out at temperature range -5~45°C	P
A.4.3	High temperature range	See below	P
A.4.3.1	General	45°C for Higher temperature range	P
A.4.3.2	Explanation of safety viewpoint	Considered	P
A.4.3.3	Safety considerations when specifying charging conditions in high temperature range	Charge at High temperature 45°C	P
A.4.3.4	Safety consideration when specifying new upper limit In high temperature range	Considered	P
A.4.4	Low temperature range	See below	P
A.4.4.1	General	0°C for Lower temperature range	P

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 30 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
A.4.4.2	Explanation of safety viewpoint	Considered	P
A.4.4.3	Safety considerations, when specifying charging conditions in low temperature range	Charge at Low temperature -5°C	P
A.4.4.4	Safety considerations when specifying a new lower limit in the low temperature range	Considered	P
A.4.5	Scope of the application of charging current	In compliance	P
A.4.6	Consideration of discharge	See below	P
A.4.6.1	General	Considered	P
A.4.6.2	Final discharge voltage and explanation of safety viewpoint	Considered	P
A.4.6.3	Discharge current and temperature range	Considered	P
A.4.6.4	Scope of application of the discharging current	Considered	P
A.5	Sample preparation	This is country specific test which is not applicable in India	N/A
A.5.1	General	As above	N/A
A.5.2	Insertion procedure for nickel particle to generate internal short	As above	N/A
A.5.3	Disassembly of charged cell	As above	N/A
A.5.4	Shape of nickel particle	As above	N/A
A.5.5	Insertion of nickel particle to cylindrical cell	As above	N/A
A.5.5.1	Insertion of nickel particle to winding core	As above	N/A
A.5.5.2	Mark the position of nickel particle on the both end of winding core of the separator	As above	N/A
A.5.6	Insertion of nickel particle to prismatic cell	As above	N/A
A.6	Experimental procedure of the forced internal short-circuit test	As above	N/A
A.6.1	Material and tools for preparation of nickel particle	As above	N/A
A.6.2	Example of a nickel particle preparation procedure	As above	N/A
A.6.3	Positioning (or placement) of a nickel particle	As above	N/A
A.6.4	Damaged separator precaution	As above	N/A
A.6.5	Caution for rewinding separator and electrode	As above	N/A

BIS_BAT/SCAB_IS16046(PART2)_V1.0



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 31 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

A.6.6	Insulation film for preventing short-circuit	As above	N/A
A.6.7	Caution when disassembling a cell	As above	N/A
A.6.8	Protective equipment for safety	As above	N/A
A.6.9	Caution in the case of fire during disassembling	As above	N/A
A.6.10	Caution for the disassembling process and pressing the electrode core	As above	N/A
A.6.11	Recommended specifications for the pressing device	As above	N/A

*- Total number of Requirements to be observed / inspected = 50

Total No of applicable Requirement = 29

No of Requirements for which the sample passed = 29

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1	IS 16046 (Part 2):2018	Page 32 of 40
Dated:12/07/2023	IEC 62133-2:2017	ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
---------	--------------------	---------------	---------

ANNEX B	RECOMMENDATIONS TO EQUIPMENT MANUFACTURERS AND BATTERY ASSEMBLERS	Information is provided by cell manufacturer	P
---------	---	--	---

ANNEX C	RECOMMENDATIONS TO THE END-USERS	Cell is used to manufacturing battery products	N/A
---------	----------------------------------	--	-----


Shweta Kumar
Senior Analyst



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 33 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

Cl. No.	Requirement + Test	Result-Remark	Verdict
ANNEX D	MEASUREMENT OF THE INTERNAL AC RESISTANCE FOR COIN CELLS		N/A
D.1	General	This is not a coin cell	N/A
D.2	Method	As above	N/A
	A sample size of three coin cells is required for this measurement	As above	N/A
	Coin cells with an internal resistance of less than or equal to 3 Ω are subjected to the testing according to Clause 6 and Table 1*	As above	N/A
	Coin cells with an internal resistance greater than 3 Ω require no further testing	As above	N/A

Total number of Requirements to be observed / inspected = 05

Total No of applicable Requirement = 00

No of Requirements for which the sample passed = N/A

Total number of tests to be conducted = 00

Total No of applicable Tests = 00

No. of tests for which the sample passed = N/A

Certificate: It is certified that the above tests were performed and found to be passing in the requirement tested.



(Approving Authority)



TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 34 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

ANNEX E	PACKAGING AND TRANSPORT	Satisfactory	P
---------	-------------------------	--------------	---

ANNEX F	COMPONENT STANDARDS REFERENCES	Applicable for battery pack	N/A
---------	--------------------------------	-----------------------------	-----

Shruti K. Kumar
Spectro Analytical Labs Pvt. Ltd.
Senior Engineer
for Analysis



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 35 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

TABLE: Critical components information

P

Object/part no.	Manufacturer/ trademark	Type/model	Technical data	Standard	Mark(s) of conformity ¹⁾
Positive Electrode	Xinxiang Hongli Power Technology Co., Ltd	HL04	Lithium Manganate (LiMn2O4)	IS 16046(Part 2): 2018/ IEC62133-2:2017	Tested with Cell
Negative Electrode	Henan Jiaozuo rongchuang graphite Technology Co., Ltd	YJ-001A	C	IS 16046(Part 2): 2018/ IEC62133-2:2017	Tested with Cell
Separator	Anhui Qianchang new energy technology Co., LTD	PE01	16µm PP PE	IS 16046(Part 2): 2018/ IEC62133-2:2017	Tested with Cell
Electrolyte	Shinghwa Advanced Material Group Co.,Ltd	SH-2304	LiPF6	IS 16046(Part 2): 2018/ IEC62133-2:2017	Tested with Cell
Cell Case	Xinxiang Shengda New Energy Technology Co., Ltd	Model:18650	18.1mm*68.5mm*0.22mm	IS 16046(Part 2): 2018/ IEC62133-2:2017	Tested with Cell

Supplementary information:

¹⁾ Evidence have been evaluated and checked for the agreed level of compliance as per the referred standard.

Shweta Kumar
Senior Analyst



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 36 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

7.2.1	TABLE: Continuous charging at constant voltage (cells)				P
Sample no.	Recommended charging voltage V _c (Vdc)	Recommended charging current I _{rec} (A)	OCV before test (Vdc)	Results	
Sample No. 01	4.15	0.4	4.2	A,B	
Sample No. 02	4.16	0.4	4.2	A,B	
Sample No. 03	4.15	0.4	4.2	A,B	
Sample No. 04	4.14	0.4	4.2	A,B	
Sample No. 05	4.15	0.4	4.2	A,B	
Supplementary information: - A: No fire or explosion - B: No leakage - C: Leakage - D: Fire - E: Explosion - F: Bulge					






TC-6314



eurofins

Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 37 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

7.3.1	TABLE: External short-circuit (cell)				P
Sample no.	Ambient T (°C)	OCV before test (Vdc)	Resistance of circuit (mΩ)	Maxlimum case temperature rise ΔT (K)	Results
Samples charged at charging temperature upper limit: 45°C					
Sample No. 06	55°C±5°C	4.15	83	44.7	A,G
Sample No. 07		4.14	86	42.4	A,G
Sample No. 08		4.16	83	43.6	A,G
Sample No. 09		4.14	87	45.8	A,G
Sample No. 10		4.15	84	44.4	A,G
Samples charged at charging temperature lower limit: -5°C					
Sample No. 11	55°C±5°C	4.15	84	44.6	A,G
Sample No. 12		4.14	85	46.5	A,G
Sample No. 13		4.16	87	45.3	A,G
Sample No. 14		4.15	85	43.4	A,G
Sample No. 15		4.14	82	44.5	A,G

Supplementary information:

- A: No fire or explosion
- B: No leakage
- C: Leakage
- D: Fire
- E: Explosion
- F: Bulge
- G: Others

Shivendra Kumar
Senior Analyst



TC-6314

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 38 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

7.3.5	TABLE: Crush (cells)				P
Sample no.		OCV before test (Vdc)	OCV at removal of crushing force (Vdc)	Maximum force applied to the cell during crush (kN)	Results
Samples charged at charging temperature upper limit: 45°C					
Sample No. 16		4.15	4.15	13.27	A
Sample No. 17		4.15	4.15	13.24	A
Sample No. 18		4.14	4.14	13.34	A
Sample No. 19		4.15	4.15	13.39	A
Sample No. 20		4.14	4.14	13.35	A
Samples charged at charging temperature lower limit: -5°C					
Sample No. 21		4.15	4.15	13.37	A
Sample No. 22		4.14	4.14	13.45	A
Sample No. 23		4.15	4.15	13.34	A
Sample No. 24		4.15	4.15	13.34	A
Sample No. 25		4.14	4.14	13.42	A
Supplementary information:					
- A: No fire or explosion					
- B: No leakage					
- C: Leakage					
- D: Fire					
- E: Explosion					
- F: Bulge					
- G: Others (please explain)					





TC-6314

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 39 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

7.3.7	TABLE: Forced discharge (cells)				P
Sample no.		OCV before application of reverse charge (Vdc)	Measured reverse charge I _t (A)	Lower limit discharge voltage (Vdc)	Results
Sample No. 26		3.17	2.0	3.0	A
Sample No. 27		3.16	2.0	3.0	A
Sample No. 28		3.18	2.0	3.0	A
Sample No. 29		3.17	2.0	3.0	A
Sample No. 30		3.17	2.0	3.0	A
Supplementary information:					
- A: No fire or explosion					
- B: No leakage					
- C: Leakage					
- D: Fire					
- E: Explosion					
- F: Bulge					
- G: Others (please explain)					

Shindea Kumar

SPECTRO ANALYTICAL LABS PRIVATE LIMITED
Shindea Kumar
Sr. Analyst



TC-6314



Spectro

Legal Entity: Spectro Analytical Labs Private Limited
(formerly Spectro Analytical Labs Limited)
(part of Eurofins Scientific SE)
CIN: U74220DL1998PTC092698

Report No.SC23EPF10473_1

IS 16046 (Part 2):2018

Page 40 of 40

Dated:12/07/2023

IEC 62133-2:2017

ULR No TC631423000007010F

PHOTOGRAPH



Cell-View


Shikendra Kumar
Senior Analyst
Spectro Analytical Labs Pvt. Ltd.